

ECON 101H, Introduction to Economics

Recitation 3: Market Equilibrium and Elasticity

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Demand:

$$Q^D = 100 - 6P$$

Supply:

$$Q^S = 10P$$

(Quantities in thousands per month)



Exercise 1: Movement Along a Curve

Suppose the price of beer rises from \$6.25 to \$7.00.

1. Does the demand curve shift?
2. Does the supply curve shift?
3. What changes: quantity demanded, quantity supplied, or both?
4. Calculate the new quantity demanded.



Exercise 2: Demand Shift Identification

For each scenario, state: (i) which curve shifts, (ii) direction.

1. A new study says moderate beer drinking is healthy.
2. The price of wine falls.
3. Student income falls in a recession.
4. UNC enrollment increases by 20%.



Exercise 3: Supply Shift Identification

For each scenario, state: (i) which curve shifts, (ii) direction.

1. Automation lowers production costs.
2. Government imposes a \$1 tax per six-pack.
3. Several breweries shut down.



Exercise 4: Elasticity of Demand

$$E_D = \frac{dQ^D}{dP} \times \frac{P}{Q}$$

1. Find $\frac{dQ^D}{dP}$.
2. Compute E_D at $P^* = 6.25$, $Q^* = 62.5$.
3. Compute E_D at $P = 10$.
4. Compute E_D at $P = 2$.
5. At what price is demand unit elastic?



Exercise 5: Supply Shift with New Equation

$$Q_2^S = 10P - 20$$

1. Find the new equilibrium using original demand.
2. Compute E_S at the new equilibrium.

